



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

**UNIVERSITI TEKNOLOGI MALAYSIA
FAKULTI SAINS**

**FINAL EXAMINATION
SEMESTER III**

COURSE CODE : FSPC 0034

COURSE NAME : CHEMISTRY II

PROGRAMME : ASASI UTMSPACE

DATE : 24th APRIL 2022

DURATION : 1 HOUR 15 MINUTES

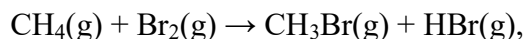
INSTRUCTION TO STUDENTS:

- 1. THE PAPER CONSISTS OF 3 QUESTIONS. ANSWER ALL QUESTIONS.**
- 2. ALL ANSWERS HAVE TO BE WRITTEN IN YOUR OWN PIECE/S OF PAPER USING BLUE/BLACK-COLOURED PEN.**
- 3. PLEASE WRITE YOUR NAME AND MATRIC CARD NUMBER ON YOUR ANSWER SHEET.**
- 4. PLEASE WRITE THE QUESTION NUMBER CLEARLY ON THE PAPER WHERE YOU ARE ANSWERING.**

(THIS BOOKLET CONSISTS OF 8 PRINTED PAGES INCLUDING THIS PAGE)

QUESTION 1 (10 MARKS)

- a. Calculate the enthalpy change of the reaction;



using the bond energies give in Table 1.

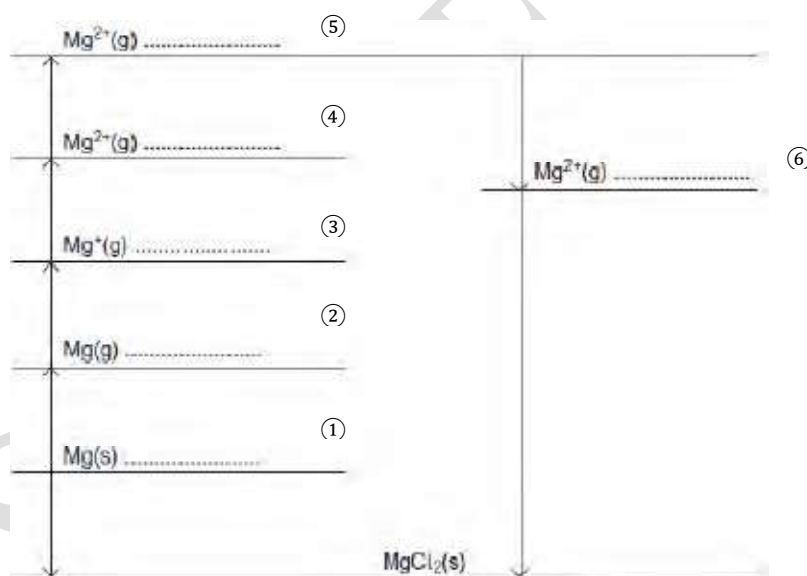
Table 1. Average bond enthalpies

bond	$\Delta H_b/\text{kJmol}^{-1}$
C-H	+413
C-Br	+280
Br-Br	+193
H-Br	+366

(5 marks)

- b. Magnesium chloride, MgCl_2 , is an ionic compound used to treat magnesium deficiency in the body.

- (i) Complete this Born–Haber cycle for magnesium chloride by giving the missing species on the dotted lines. Include state symbols where appropriate. The energy levels are not drawn to scale.



(3 marks)

- (ii) Use your Born–Haber cycle from part (i) and data from Table 2 to calculate the value for the electron affinity of chlorine.

Table 2. Enthalpy change in a Born-Haber cycle.

Enthalpy Change	Enthalpies ($\Delta H/\text{kJmol}^{-1}$)
Enthalpy of atomisation of magnesium	+150
Enthalpy of atomisation of chlorine	+121
First ionisation energy of magnesium	+736
Second ionisation energy of magnesium	+1450
Enthalpy of formation of magnesium chloride	-642
Lattice enthalpy of formation of magnesium chloride	-2493

(2 marks)

QUESTION 2 (10 MARKS)

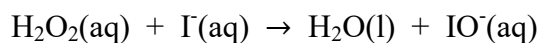
- a. A flask is charged with 0.100 M of **A** and allowed to decompose to form **B** according to the hypothetical gas-phase reaction **A (g) → B (g)**. The following data are collected:

Time (s)	0	40	80	120	160
Molarity of A (M)	0.100	0.091	0.083	0.077	0.071

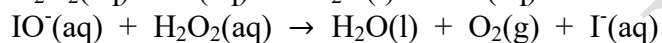
- Calculate the average rate of disappearance of A from 40 s to 120 s time.
- Determine the order of this reaction
- Determine the rate of reaction at 60 s.

(1+3+4 marks)

- b. Given a two step reaction equations:



SLOW



FAST

- State the intermediate in the reaction
- Determine the rate law of the reaction

(1+1 marks)

2) (a) (i)

$$k = \frac{1}{a} x - \frac{d[A]}{dt}$$

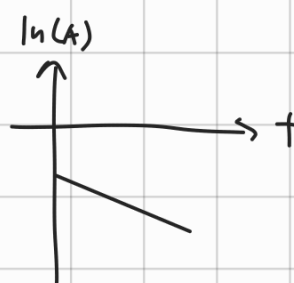
$$= \frac{1}{1} x - \frac{0.077 - 0.091}{120 - 40}$$

$$= 1.75 \times 10^{-4} \text{ ms}^{-1}$$

(ii)

$\ln(A)$	$\frac{1}{A}$	t
-2.303	10	0
-2.396	11	40
-2.489	12	80
-2.564	13	120
-2.645	14	160

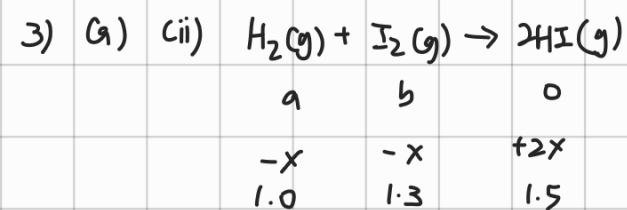
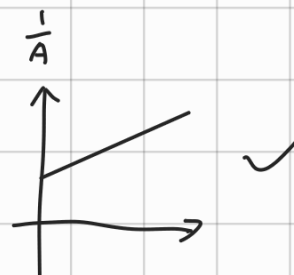
$\therefore$ Second order



(b) (i) rate = $[H_2O_2][I^-]$

$\therefore$ second order

(ii) IO^- is intermediate



$$2x = 1.5 \text{ M}$$

$$x = 0.75 \text{ M}$$

$$a = 1.75 \text{ M}$$

$$b = 2.05 \text{ M}$$

$$Q_c = \frac{0}{0.75 \times 2.05}$$

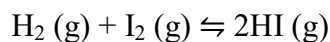
$$= 0$$

(iii) $Q_c = 0, K_c = 20$

$Q_c < K_c$, hence not equilibrium, shift to right

QUESTION 3 (10 MARKS)

$$(9) (i) \quad Q_c = \frac{[HI]^2}{[H_2][I_2]}$$



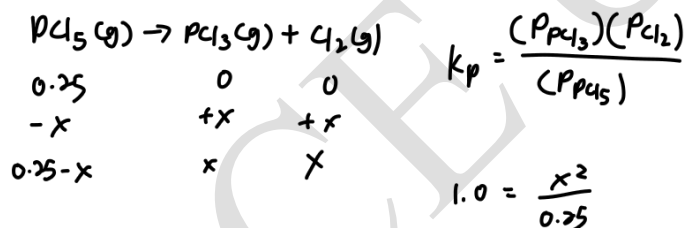
- a. The equilibrium constant, K_c for the above reaction at 200 °C is 20.0. At the point during the reaction, a mixture containing 1.0 mol dm⁻³ of H_2 , 1.3 mol dm⁻³ of I_2 and 1.5 mol dm⁻³ of HI is left standing at 200 °C.

- Write the reactions quotient, Q_c for the reaction.
- Calculate the reaction quotient, Q_c for the reaction.
- Determine whether the reaction is in the equilibrium or not. Then, determine the net direction the reaction is progressing.

(1+2+2 marks)

- b. The equilibrium constant, K_p for the decomposition of phosphorus pentachloride gas to phosphorus trichloride gas and chlorine gas is 1.00 at 250 °C.

- Write down the balanced chemical equation including their physical state and the equilibrium constant in terms of K_p for this reaction.
- Calculate the partial pressures of all reacting species at the equilibrium if the initial pressure partial pressure of phosphorus pentachloride is 0.25 atm flask at 250 °C.



(2+3 marks)

$$P_{PCl_5} = 0.20 \text{ atm}$$

$$P_{PCl_3} = 0.05 \text{ atm}$$

$$P_{Cl_2} = 0.05 \text{ atm}$$

$$x = 0.05 \text{ atm}$$

1																	18	
IA																	VIIIA	
1																	2	
H	2																	He
1.01	IIA																	4.00
3	4																	10
Li	Be																	Ne
6.94	9.01																	20.18
11	12																	18
Na	Mg																	Ar
22.99	24.31																	39.95
19	20																	36
K	Ca																	Kr
39.1	40.08																	83.80
37	38																	54
Rb	Sr																	Xe
85.47	87.62																	131.29
55	56																	86
Cs	Ba																	
132.9	137.3																	
87	88																	
Fr	Ra																	
(223)	(226)																	
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LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$
		=	$2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$
		=	$0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	μ_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	β_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$

STANDARD REDUCTION POTENTIAL (E°_{SRP})

Half-Reaction	E°_{red}
$\text{K}^+ + \text{e}^- \rightleftharpoons \text{K}$	-2.924
$\text{Ba}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ba}$	-2.900
$\text{Ca}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ca}$	-2.760
$\text{Na}^+ + \text{e}^- \rightleftharpoons \text{Na}$	-2.712
$\text{Mg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mg}$	-2.375
$\text{H}_2 + 2\text{e}^- \rightleftharpoons 2\text{H}^-$	-2.230
$\text{Al}^{3+} + 3\text{e}^- \rightleftharpoons \text{Al}$	-1.706
$\text{Mn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Mn}$	-1.040
$\text{Zn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Zn}$	-0.763
$\text{Cr}^{3+} + 3\text{e}^- \rightleftharpoons \text{Cr}$	-0.740
$\text{S} + 2\text{e}^- \rightleftharpoons \text{S}^{2-}$	-0.508
$2\text{CO}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{C}_2\text{O}_4$	-0.490
$\text{Cr}^{3+} + \text{e}^- \rightleftharpoons \text{Cr}^{2+}$	-0.410
$\text{Fe}^{2+} + 2\text{e}^- \rightleftharpoons \text{Fe}$	-0.409
$\text{Co}^{2+} + 2\text{e}^- \rightleftharpoons \text{Co}$	-0.280
$\text{Ni}^{2+} + 2\text{e}^- \rightleftharpoons \text{Ni}$	-0.230
$\text{Sn}^{2+} + 2\text{e}^- \rightleftharpoons \text{Sn}$	-0.136
$\text{Pb}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pb}$	-0.126
$\text{Fe}^{3+} + 3\text{e}^- \rightleftharpoons \text{Fe}$	-0.036
$2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2$	0.000
$\text{S}_4\text{O}_6^{2-} + 2\text{e}^- \rightleftharpoons 2\text{S}_2\text{O}_3^{2-}$	0.089
$\text{Sn}^{4+} + 2\text{e}^- \rightleftharpoons \text{Sn}^{2+}$	0.150
$\text{Cu}^{2+} + \text{e}^- \rightleftharpoons \text{Cu}^+$	0.158
$\text{Cu}^{2+} + 2\text{e}^- \rightleftharpoons \text{Cu}$	0.340
$\text{O}_2 + 2\text{H}_2\text{O} + 4\text{e}^- \rightleftharpoons 4\text{OH}^-$	0.401
$\text{Cu}^+ + \text{e}^- \rightleftharpoons \text{Cu}$	0.522
$\text{I}_3^- + 2\text{e}^- \rightleftharpoons 3\text{I}^-$	0.534
$\text{MnO}_4^- + 2\text{H}_2\text{O} + 3\text{e}^- \rightleftharpoons \text{MnO}_2 + 4\text{OH}^-$	0.588
$\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{H}_2\text{O}_2$	0.682
$\text{Fe}^{3+} + \text{e}^- \rightleftharpoons \text{Fe}^{2+}$	0.770
$\text{Hg}_2^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.796
$\text{Ag}^+ + \text{e}^- \rightleftharpoons \text{Ag}$	0.800
$\text{Hg}^{2+} + 2\text{e}^- \rightleftharpoons \text{Hg}$	0.851
$\text{H}_2\text{O}_2 + 2\text{e}^- \rightleftharpoons 2\text{OH}^-$	0.880
$\text{HNO}_3 + 3\text{H}^+ + 3\text{e}^- \rightleftharpoons \text{NO} + 2\text{H}_2\text{O}$	0.960
$\text{Br}_2(\text{aq}) + 2\text{e}^- \rightleftharpoons 2\text{Br}^-$	1.087
$2\text{IO}_3^- + 12\text{H}^+ + 10\text{e}^- \rightleftharpoons \text{I}_2 + 6\text{H}_2\text{O}$	1.190
$\text{CrO}_4^{2-} + 8\text{H}^+ + 3\text{e}^- \rightleftharpoons \text{Cr}^{3+} + 4\text{H}_2\text{O}$	1.195
$\text{Pt}^{2+} + 2\text{e}^- \rightleftharpoons \text{Pt}$	1.200
$\text{MnO}_2 + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 2\text{H}_2\text{O}$	1.208
$\text{O}_2 + 4\text{H}^+ + 4\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	1.229
$\text{Cr}_2\text{O}_7^{2-} + 14\text{H}^+ + 6\text{e}^- \rightleftharpoons 2\text{Cr}^{3+} + 7\text{H}_2\text{O}$	1.330
$\text{Cl}_2(\text{g}) + 2\text{e}^- \rightleftharpoons 2\text{Cl}^-$	1.358
$\text{PbO}_2 + 4\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{Pb}^{2+} + 2\text{H}_2\text{O}$	1.467
$\text{MnO}_4^- + 8\text{H}^+ + 5\text{e}^- \rightleftharpoons \text{Mn}^{2+} + 4\text{H}_2\text{O}$	1.491
$\text{Au}^+ + \text{e}^- \rightleftharpoons \text{Au}$	1.680
$\text{H}_2\text{O}_2 + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{H}_2\text{O}$	1.776
$\text{Co}^{3+} + \text{e}^- \rightleftharpoons \text{Co}^{2+}$	1.842
$\text{S}_2\text{O}_8^{2-} + 2\text{e}^- \rightleftharpoons 2\text{SO}_4^{2-}$	2.050
$\text{O}_3(\text{g}) + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons \text{O}_2(\text{g}) + \text{H}_2\text{O}$	2.070
$\text{F}_2(\text{g}) + 2\text{H}^+ + 2\text{e}^- \rightleftharpoons 2\text{HF}(\text{aq})$	3.030